

Module 08: Planning in Embodied

Learning Objectives

1. Define **Planning** within the context of Embodied.
2. Analyze how Planning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Embodied to the formal definition of Planning.

Introduction

This module explores **Planning**. In the **Embodied** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Planning is a critical component of the 8-part Active Inference spine, bridging the gap between Communication and Systems.

Key Concepts

1. Planning as a Markov Blanket Boundary

How does Planning define the boundary between the agent and the environment?

2. Generative Models of Planning

What parameters involved in Planning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Planning drive the perception-action loop?

Applications

In Embodied, we see Planning manifest in: * **Specific Example 1:** A veteran military commander surveys a battlefield and intuitively selects a strategy without formal analysis,

feeling a bodily sense of rightness about one approach and unease about another; this is intuitive planning, where the generative model runs compressed counterfactual simulations drawing on deep experiential priors, and the expected free energy of each possible trajectory is evaluated somatically -- the strategy that produces the least visceral tension is the one the body's predictive model identifies as optimal. * **Specific Example 2:** An experienced entrepreneur considering two investment opportunities reports that one "feels alive" in her body while the other produces a dull heaviness despite similar financial projections; this gut-level planning reflects the generative model's capacity to integrate vast amounts of implicit pattern knowledge -- market dynamics, team chemistry, timing -- into a single interoceptive signal that functions as a somatic summary statistic of expected long-term free energy, guiding strategic choice through embodied intuition rather than spreadsheet analysis.

Conclusion

Understanding Planning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.